

7.15 Block 00b audit — review candidate

Status: **local preflight complete; final report-only head awaiting hosted qualification.** This report records the code-bearing 00b reconciliation on the accepted 00a foundation. It is an audit-unit receipt, not release, merge, or hardware approval.

1. Identity and frozen scope

- Owner: Codex /root, vault worktree keepkey-vault-v11-agent-3; claim: docs/release/audit-claims/715-00b/OWNER.md in the main firmware worktree.
- Main-worktree master template SHA-256 at intake: fbd340efde3c2e8ec0c0932c17e10b8c3b1592f3f25eb82562467d3ac7ae6d61.
- Accepted predecessor: Block 00a PR #844, branch release/715-stack-00-audit-base, head 9cb72384fd14b2a688a70d17711a7923941d2290, tree 8f3be92b55df9a5582f4fb4669bad65416ae16ae.
- Original code-bearing Block 00b PR #845, branch release/715-stack-00b-release-foundation, head 4c56e19e3070772a28b2890a4ec862ecf77d5429, targets stale base a9173b32eb8515db8130f0bd393bc9b249d83320.
- Reconciled branch audit/715-00b-reconciled in /private/tmp/kk-715-00b-reconciled replays the original 00b source commit as fa971c617e16612be0af67ca91a0838b98f38544 on the accepted 00a predecessor. Storage and display remediation is 8e515907a2ab5fbeb2ff2956965b3110edfc040f. The C-2 boundary commit is 87a02fe8cfddec930358fd8d37ea00409e214d977. A code-bearing PR against the accepted 00a branch is pending.
- 00b adds or changes chain handlers (Hive, TRON, Zcash, Solana, THOR/Maya), signed metadata, BIP-85, recovery and storage behavior, transport/emulator support, native tests, and build configuration. This review covers the adjacent diff, not later stacks or physical hardware acceptance.

Dependency	Reconciled pin	Provenance / status
deps/python-keepkey	98c717ff2204124bf67cc78cab8ca1d501fd4e92	Accepted 00a pin; descendant of original 00b pin.
deps/device-protocol	dd9c85dc747cf965fb0e7bf49615dc9e7568ee65	Exact reconciled pin; generated-wire compatibility is exercised by the build and Python matrix.
deps/crypto/trezor-firmware	74908938c562ec89dcae9fb4e3b05ec3ff17f58b	Descendant of 00a and 00b pins; upstream PR #12 is not yet merged.

2. Git inventory and plain-language changes

The original #845 adjacent diff contains 104 paths and 16,831 added / 1,585 deleted lines. It is code-bearing. Its base predates accepted 00a, so a review there cannot establish the current dependency interaction. The reconciled adjacent diff was 102 paths and 16,824 added / 1,584 deleted lines before remediation. The first report-bearing head f7cd316a499fbccdfbcd0c532bffb6365e19474 has 108 paths, 17,173 added and 1,581 deleted lines against accepted 00a. This report-only follow-up changes the source/PDF counts; the PR body records the final exact adjacent diff. Submodule pointer changes count as one added and one deleted line each; the dependencies themselves are separate repositories. Historical counts must not be summed as the net candidate diff.

The large added groups are Hive and Zcash handlers/tests, Solana handler and test changes, signed metadata and its tests, TRON support, emulator and Python harness changes, plus smaller storage, recovery, signing, UI, and build edits. The current candidate preserves 00a's Python pin, emulator image arguments, unit-test failure propagation, and the 00a selector-bound comment while adding 00b's functionality. The reconciled crypto pin preserves bounded Base32/Base58 operations and RedPallas fixes from both predecessors.

C-2 bootloader boundary

The owner retained “No bootloader changes” as a hard 7.15 release constraint. The direct bootloader source file now matches pre-foundation commit fc1e93746132553ad98ed60f4847c8d770732bf9; the shared signatures_ok() implementation matches accepted 00a. The F3 double-hash and infective signature-verification change is withdrawn from 00b, since that shared function is called by the bootloader. A manual opt-in build still produces a different bootloader binary from the pre-foundation build (2142a51c... versus a831baf0..., SHA-256) because other shared libraries have changed. Therefore the default release build

disables bootloader, updater and bootstrap targets. Clean full and Bitcoin-only ARM builds produced none of those artifacts; CI now fails if any are present. The opt-in build is outside the 7.15 release and must not be packaged as its bootloader.

3. Current-source reconciliation and findings

ID	Finding and impact	Disposition / evidence
00b-01	The original 00b PR is based before accepted 00a. A clean review of #845 would omit real interaction changes.	Fixed in reconciled code-bearing branch based on exact accepted 00a head. Original #845 needs an explicit supersession note; new PR pending.
00b-02	<code>storage_commit()</code> serialized before setting the stor marker; freshly created wallets were invisible on reboot. #845 CI failed five power-cycle tests in both variants.	Marker moved before serialization. Five focused tests pass on the reconciled source.
00b-03	Accepted 00a Python pin recognizes legacy unframed V17 only with erased trailer bytes; the 00b writer zero-filled them.	Writer leaves the eight post-payload bytes erased. Five focused power-cycle/migration tests pass. The writer still emits the legacy format; CRC framing is separate format work.
00b-04	Opaque metadata fields displayed only their first 16 bytes before raw-data confirmation was suppressed.	Candidate displays every byte on numbered 16-byte pages. Both ARM builds and full/Bitcoin-only native, OLED and integration suites pass.
00b-05	<code>bn_format()</code> can return zero and clear output for a value that exceeds the 48-byte buffer; code ignored this and could show an empty amount.	Candidate uses 96 bytes and refuses approval if formatting or final screen assembly cannot fit. Both ARM builds and full/Bitcoin-only native, OLED and integration suites pass.
00b-06	00b changed <code>lib/board/signatures.c</code> , which the bootloader links and calls; 00a also edited <code>tools/bootloader/usb_flash.c</code> . SRS C-2 says no bootloader changes.	Owner explicitly retained C-2 and required removal. Candidate restores the pre-foundation bootloader source, drops 00b's shared F3 signature hardening, and makes bootloader/updater/bootstrap builds opt-in. A manual opt-in bootloader build still differs from the pre-foundation binary because of other shared-library changes; the default release must emit no bootloader artifacts. Clean-build proof pending. F3 hardening is withdrawn from this release scope.
00b-07	Crypto pin comes from upstream PR #12, which is not merged.	External publication dependency remains open; pin ancestry was verified.
00b-08	The Bitcoin-only Compose overlay still tagged <code>firmware-unit</code> as the full emulator image. A supposed Bitcoin-only native run executed 217 full firmware tests, exposing the mismatch.	Overlay gives both <code>firmware-unit</code> and Python tests distinct Bitcoin-only image tags. The corrected run executed 95 firmware, 20 board and 19 crypto tests; Bitcoin-only OLED and integration passed.

4. Verification and artifact identity

Check	Candidate / command	Result and limits
Focused storage power cycle	Reconciled Docker image; <code>TestStorageUpgradePrese rvation</code>	5 passed after marker and trailer fixes.
Full native	C-2 Docker image; <code>firmware-unit</code>	217 firmware, 20 board, 19 crypto passed.
Full Python/OLED	C-2 Docker Compose, declared capability flags	64 passed/23 skipped OLED; 559 passed/272 skipped integration.
Bitcoin-only native	Corrected overlay, distinct image tag	95 firmware, 20 board, 19 crypto passed.
Bitcoin-only Python/OLED	Corrected overlay, declared capability flags	25 passed/62 skipped OLED; 334 passed/497 skipped integration.
ARM/SRAM	Clean full and Bitcoin-only <code>MinSizeRel</code> builds	Both pass. SRAM reserve 21,736 B full, 31,368 B Bitcoin only; largest frame 7,664 B. Neither build emits bootloader/updater/bootstrap outputs.
Hosted CI	<code>Run 35921519743</code> on <code>f7cd316a499fbccdcbcd0c532bffb6365e19474</code>	All required jobs passed, including both ARM variants, native suites, Python integration, report and release gates. Emulator publication was intentionally skipped. This report-only follow-up head needs its own exact-head run, attached to the PR after push.

The full ARM firmware binary SHA-256 is 1c1fb14cfa1e41171279543257e7492dc5f56d820a13d826d3ad1267280c0928; the Bitcoin-only firmware binary SHA-256 is 7ad9bdbac151182a24a05aabdda412378cee82a6c766cc0091ea2996e9738fd9. Both were produced from clean out-of-tree builds of the C-2 code commit. actionlint passes. The local code-review diff gate recognizes implementation, test, dependency, build and workflow paths against the accepted 00a base.

Capability skips are intentional only when recorded in the release matrix; they do not count as a passing assertion. Emulator tests are not a substitute for physical signing, OLED, power interruption, or signed-device upgrade checks. Upstream crypto PR #12 remains a publication dependency. C-2 is enforced by excluding bootloader family artifacts from default release builds.

5. Report render and local preflight

- [x] Source and PDF include the frozen pre-report code inventory and findings.
- [x] PDF pages opened and checked for legibility.
- [x] Live base is accepted 00a and the adjacent PR diff is code-bearing.
- [x] git diff --check, code-review diff gate, exact pins and artifact IDs recorded.
- [] Full, Bitcoin-only, ARM/SRAM, and hosted exact-head results reconciled; all local and first hosted matrices pass, final report-only CI pending.

6. Copilot checkpoint

No Copilot request has been made. The authorized late review belongs on the final code-bearing adjacent diff after the checks and preflight above. A clean review of a documentation-only receipt or stale #845 base would not qualify this block.

Appendix: pre-report adjacent path inventory

This inventory is frozen at code commit 87a02fe8cfddec930358fd8d37ea00409e214d977 before adding this report and its PDF. Final preflight accounts for those report artifacts separately.

Path	Added	Deleted
.github/workflows/ci.yml	16	0
.gitmodules	2	2
CMakeLists.txt	26	1
cmake/toolchains/mingw-w64-x86_64.cmake	40	0
deps/crypto/CMakeLists.txt	11	0
deps/crypto/trezor-firmware	1	1
deps/device-protocol	1	1
docs/release/audit-units/715-00b-storage-persistence.md	88	0
include/keepkey/board/bsd_compat.h	28	0
include/keepkey/board/draw.h	25	0
include/keepkey/board/layout.h	18	0
include/keepkey/board/messages.h	6	0
include/keepkey/emulator/libkkemu.h	61	17
include/keepkey/firmware/app_confirm.h	3	0
include/keepkey/firmware/app_layout.h	9	1
include/keepkey/firmware/bip85.h	22	0
include/keepkey/firmware/eip712.h	2	9
include/keepkey/firmware/ethereum.h	1	0
include/keepkey/firmware/ethereum_contracts/thortx.h	27	1
include/keepkey/firmware/ethereum_tokens.h	4	0
include/keepkey/firmware/fsm.h	20	0

Path	Added	Deleted
include/keepkey/firmware/hive.h	255	0
include/keepkey/firmware/mayachain.h	9	0
include/keepkey/firmware/osmosis.h	16	0
include/keepkey/firmware/signed_metadata.h	240	0
include/keepkey/firmware/solana.h	151	8
include/keepkey/firmware/storage.h	7	0
include/keepkey/firmware/tendermint.h	4	0
include/keepkey/firmware/thorchain.h	19	1
include/keepkey/firmware/tron.h	55	0
include/keepkey/firmware/zcash.h	384	0
include/keepkey/transport/interface.h	2	0
include/keepkey/transport/messages-ethereum.options	25	0
include/keepkey/transport/messages-hive.options	58	0
include/keepkey/transport/messages-solana.options	4	0
include/keepkey/transport/messages-zcash.options	54	0
lib/board/draw.c	82	3
lib/board/layout.c	162	3
lib/board/messages.c	90	40
lib/board/timer.c	28	2
lib/board/udp.c	6	0
lib/board/usb.c	29	20
lib/emulator/CMakeLists.txt	7	0
lib/emulator/libkemu.c	261	26
lib/emulator/setup.c	31	2
lib/emulator/udp.c	8	2
lib/firmware/CMakeLists.txt	1	0
lib/firmware/app_confirm.c	25	0
lib/firmware/app_layout.c	89	3
lib/firmware/bip85.c	105	0
lib/firmware/eip712.c	8	1
lib/firmware/eos.c	6	0
lib/firmware/ethereum.c	128	33
lib/firmware/ethereum_contracts.c	2	0
lib/firmware/ethereum_contracts/thortx.c	169	172
lib/firmware/ethereum_contracts/zxapliquid.c	2	1
lib/firmware/ethereum_contracts/zxliquidtx.c	11	6
lib/firmware/fsm.c	15	0
lib/firmware/fsm_msg_bip85.h	142	0
lib/firmware/fsm_msg_common.h	11	0
lib/firmware/fsm_msg_ethereum.h	138	0
lib/firmware/fsm_msg_hive.h	991	0
lib/firmware/fsm_msg_mayachain.h	30	1
lib/firmware/fsm_msg_ripple.h	15	0
lib/firmware/fsm_msg_solana.h	655	165

Path	Added	Deleted
lib/firmware/fsm_msg_thorchain.h	23	2
lib/firmware/fsm_msg_tron.h	185	79
lib/firmware/fsm_msg_zcash.h	1414	0
lib/firmware/hive.c	1016	0
lib/firmware/mayachain.c	91	77
lib/firmware/messagemap.def	6	0
lib/firmware/osmosis.c	18	0
lib/firmware/recovery_cipher.c	59	5
lib/firmware/reset.c	21	7
lib/firmware/ripple.c	19	0
lib/firmware/signed_metadata.c	1064	0
lib/firmware/solana.c	568	291
lib/firmware/storage.c	65	7
lib/firmware/storage.h	5	0
lib/firmware/tendermint.c	28	0
lib/firmware/thorchain.c	37	7
lib/firmware/tiny-json.c	5	3
lib/firmware/tron.c	359	0
lib/firmware/zcash.c	1133	0
lib/rand/rng.c	10	0
lib/transport/CMakeLists.txt	18	0
scripts/build/docker/device/release.sh	6	1
scripts/emulator/docker-compose.bitcoin-only.yml	2	0
scripts/emulator/docker-compose.yml	9	0
scripts/emulator/firmware-unit.sh	8	0
scripts/generate-test-report.py	71	6
tools/CMakeLists.txt	5	3
tools/bootloader/usb_flash.c	4	0
tools/emulator/CMakeLists.txt	33	11
tools/firmware/CMakeLists.txt	3	1
unittests/crypto/CMakeLists.txt	16	0
unittests/firmware/coins.cpp	11	0
unittests/firmware/ethereum.cpp	1	32
unittests/firmware/hive.cpp	783	0
unittests/firmware/mayachain.cpp	7	0
unittests/firmware/signed_metadata.cpp	1551	0
unittests/firmware/solana.cpp	905	505
unittests/firmware/storage.cpp	107	4
unittests/firmware/thorchain.cpp	74	18
unittests/firmware/tron.cpp	495	0
unittests/firmware/zcash.cpp	1757	0